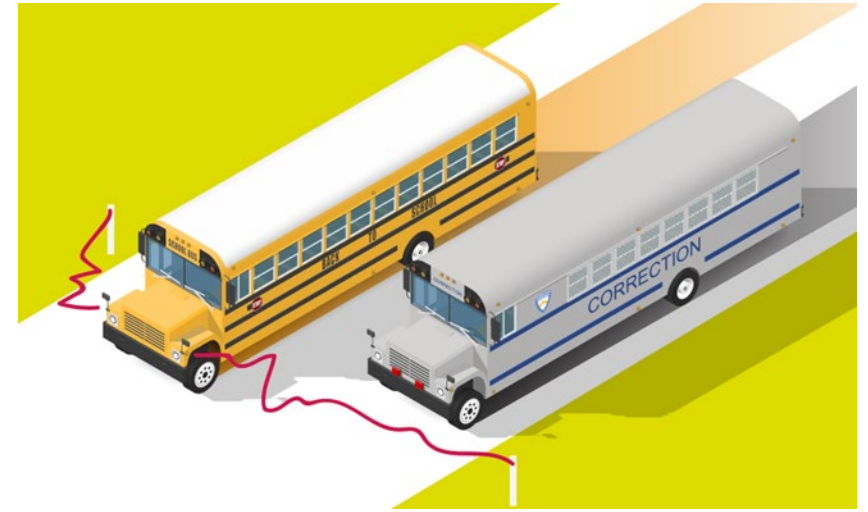


# Fuzzy Regression Methods & Alternatives



**Geek Seminar: Queensland University of Technology**  
**February 27, 2026**

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**BITA**

ARC Training Centre for Behavioural  
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# Fuzzy Logic (Stanford Encyclopedia of Philosophy)

## What is Fuzzy Logic:

Fuzzy logic is intended to model **logical reasoning** with vague or imprecise statements like “Petr is young (rich, tall, hungry, etc.)”. It refers to a family of [many-valued logics](#), where the [truth-values](#) are interpreted as **degrees of truth**. The truth-value of a logically compound proposition, like “Carles is tall and Chris is rich”, is determined by the truth-value of its components. In other words, like in classical logic, one **imposes** truth-functionality

## Where did it come from:

Fuzzy logic emerged in the context of the theory of fuzzy sets, introduced by Lotfi Zadeh (1965). A fuzzy set assigns a degree of membership, typically a real number from the interval  $[0, 1]$ , to elements of a universe. Fuzzy logic arises by assigning degrees of truth to propositions. The standard set of truth-values (degrees) is the real unit interval  $[0, 1]$ , where 0 represents “totally false”, 1 represents “totally true”, and the other values refer to partial truth, i.e., intermediate degrees of truth.<sup>[1]</sup>

See [Stanford Encyclopedia of Philosophy](#) for mathematical proofs

# Fuzzy Logic (Stanford Encyclopedia of Philosophy)

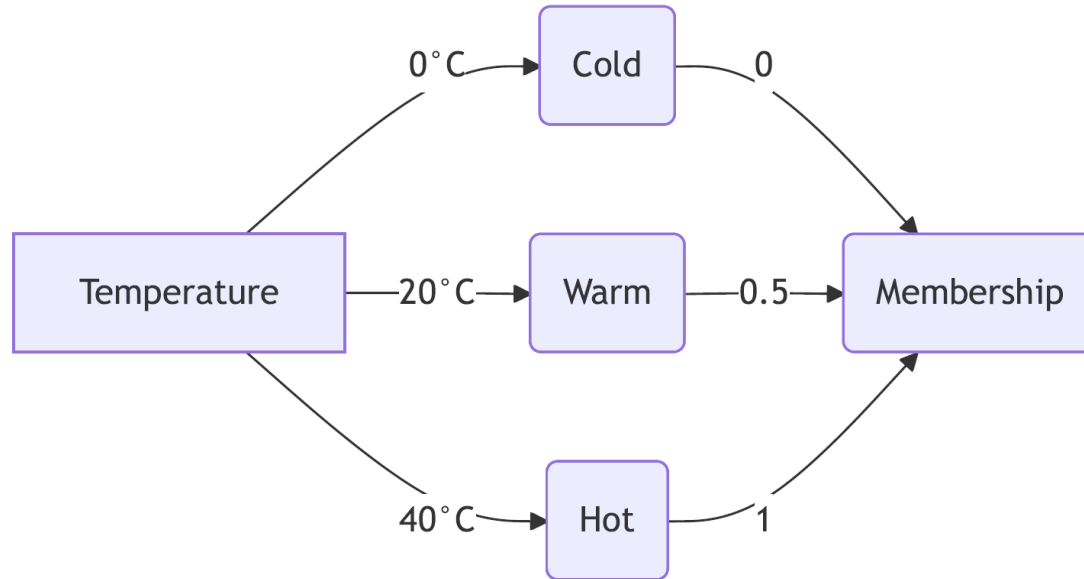
## **“Systematic handling of degrees of some kind”**

“Fuzzy logic” is often understood in a very wide sense which includes all kinds of formalisms and techniques referring to the systematic handling of degrees of some kind (see, e.g., Nguyen & Walker 2000). In particular in engineering contexts (fuzzy control, fuzzy classification, soft computing) it is aimed at **efficient computational methods** tolerant to suboptimality and imprecision (see, e.g., Ross 2010). This entry focuses on fuzzy logic in a restricted sense, established as a discipline of mathematical logic following the seminal monograph by Petr Hájek (1998) and nowadays usually referred to as “mathematical fuzzy logic”. For details about the history of different variants of fuzzy logic we refer to Bělohlávek, Dauben, & Klir 2017.

## **Focus:**

Mathematical fuzzy logic focuses on logics based on a **truth-functional account of partial truth** and studies them in the spirit of classical mathematical logic, investigating syntax, model-theoretic semantics, proof systems, completeness, etc., both, at the propositional and the predicate level (see Cintula, Fermüller, Hájek, & Noguera 2011 and 2015).

# Simple Example : define a fuzzy set "high temperature"



The membership function is typically represented using a mathematical expression, such as a triangular or trapezoidal function. For instance, a triangular membership function can be defined as:

$$\mu(x) = \begin{cases} 0 & x < a \\ \frac{x - a}{b - a} & a \leq x < b \\ \frac{c - x}{c - b} & b \leq x < c \\ 0 & x \geq c \end{cases}$$

where  $a$ ,  $b$ , and  $c$  are parameters that define the shape of the membership function.

# Method

- **Fuzzy regression methods** are extensions of classical regression techniques designed to handle uncertainty, vagueness, and imprecision in data

The method establishes a fuzzy functional relationship between dependent and independent variables, where the input data can be either crisp or fuzzy.

It is widely used in situations where the information is partly true: e.g. forecasting, decision-making, control systems, medical diagnosis and modeling when data is incomplete, noisy, or linguistically (AI) described.

(Chat GPT 17 February 2026)

# Primary approaches to fuzzy regression

## **Tanaka's Fuzzy Linear Regression**

(FLR) is a method that incorporates fuzzy logic principles to handle uncertainty in regression analysis. It is particularly useful when dealing with data that is imprecise or uncertain. The method involves the following steps:

**Data Preparation:** The data is prepared by converting it into fuzzy numbers, which represent the uncertainty associated with the data.

**Modeling:** The regression model is constructed using the fuzzy numbers, taking into account the uncertainty.

**Prediction:** The model is used to make predictions based on the fuzzy numbers, which can then be interpreted as fuzzy predictions.

## **Application:**

Tanaka's FLR is widely used in various fields, including risk assessment, decision-making, and quality control. It provides a systematic approach to handling uncertainty in regression analysis, making it a valuable tool for researchers and practitioners alike.

# Applications

- **Risk Assessment:** For example, fuzzy linear regression models have been used to assess the risks of cumulative trauma disorders.

Shapiro, A. F., & Koissi, M. C. (2015). Risk assessment applications of fuzzy logic. *Casualty Actuarial Society, Canadian Institute of Actuaries, Society of Actuaries*, 390.

- **Forecasting:** It has been applied to forecast in uncertain environments, such as using fuzzy ARIMA models for predicting foreign exchange markets.

Kuo, R. J., & Xue, K. C. (1999). Fuzzy neural networks with application to sales forecasting. *Fuzzy Sets and Systems*, 108(2), 123-143.

- **Industrial Engineering:** Fuzzy regression models have been used for manpower forecasting and modeling transfer molding for microchip encapsulation.

Kahraman, C. (Ed.). (2006). *Fuzzy applications in industrial engineering* (Vol. 201). Heidelberg: Springer.

# Applications

- **Economics:** “Cattaneo, Idrobo & Titiunik’s (2024) monograph, together with its accompanying first part (Cattaneo, Idrobo, and Titiunik, 2020), collects and expands the instructional materials we prepared for more than 50 short courses and workshops on Regression Discontinuity (RD) methodology that we taught between 2014 and 2023.”

*Cattaneo, M. D., N. Idrobo, and R. Titiunik (2020): A Practical Introduction to Regression Discontinuity Designs: Foundations. Cambridge Elements: Quantitative and Computational Methods for Social Science, Cambridge University Press.*

*Cattaneo, M. D., Idrobo, N., & Titiunik, R. (2024). A Practical Introduction to Regression Discontinuity Designs: Extensions [Google-Books-ID: 9zYAEQAAQBAJ].*

*Battistin, E., & Rettore, E. (2008). Ineligibles and eligible non-participants as a double comparison group in regression-discontinuity designs. Journal of Econometrics, 142(2), 715-730.*

- **Example Usage:**

*Beatton, T., Kidd, M. P., Niu, A., & Vella, F. (2023). Age of Starting School, Academic Performance, and the Impact of Non-Compliance: An Experiment within an Experiment, Evidence from Australia. Economic Record, 99(325), 175-206.*

*Beatton, T., Kidd, M.P. & Sandi, M. (2022). School Indiscipline and Crime. CESifo Working Papers, 9526. (Revise and resubmit, Economics of Education Review).*



# Primary approaches to fuzzy regression

- **Tanaka's Fuzzy Linear Programming Approach:** This method uses linear programming to optimize a functional that measures the spreads of the estimated output. It can handle both linear and exponential possibilistic regression models.

Tanaka, H., Uejima, S., Asai, K., Fuzzy linear regression model, IEEE Transactions on Systems Man Cybernetics, Vol. 12, pp. 903-907, 1982.

- “In the usual regression model, **deviations between the observed values and the estimated values** are supposed to be due to **measurement errors**. Here, on the contrary, these deviations are assumed to depend on the **indefiniteness of the system structure ... fuzziness of the parameters of the system.**”

# Tanaka's FLP method: House Price Example

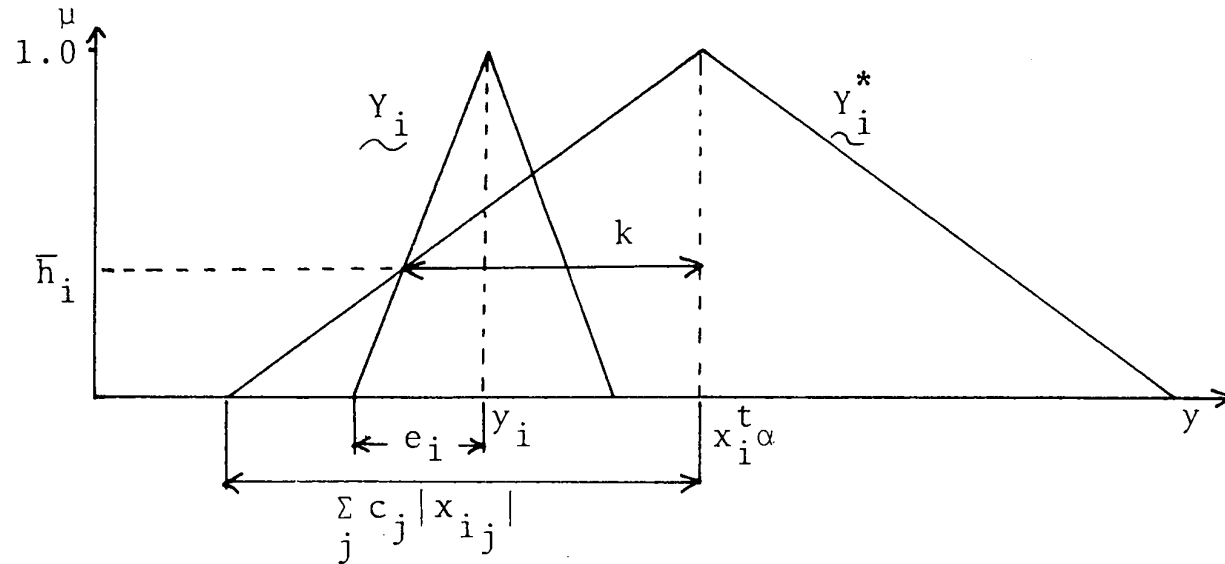


Fig. 4. Degree of fitting of  $\tilde{Y}_i^*$  to given fuzzy data  $\tilde{Y}_i$ .

[Tanaka, H., Uejima, S., Asai, K., Fuzzy linear regression model, IEEE Transactions on Systems Man Cybernetics, Vol. 12, pp. 903-907, 1982.](#)

# Tanaka's FLP House Price Example

TABLE III  
INPUT-OUTPUT DATA CONCERNING HOUSE PRICES

| NO. | $\tilde{Y}=(y, e)$ | $x_0$ | $x_1$ | $x_2$  | $x_3$ | $x_4$ | $x_5$ |
|-----|--------------------|-------|-------|--------|-------|-------|-------|
| 1   | (6060 , 550)       | 1     | 1     | 38.09  | 36.43 | 5     | 1     |
| 2   | (7100 , 50)        | 1     | 1     | 62.10  | 26.50 | 6     | 1     |
| 3   | (8080 , 400)       | 1     | 1     | 63.76  | 44.71 | 7     | 1     |
| 4   | (8260 , 150)       | 1     | 1     | 74.52  | 38.09 | 8     | 1     |
| 5   | (8650 , 750)       | 1     | 1     | 75.38  | 41.40 | 7     | 2     |
| 6   | (8520 , 450)       | 1     | 2     | 52.99  | 26.49 | 4     | 2     |
| 7   | (9170 , 700)       | 1     | 2     | 62.93  | 26.49 | 5     | 2     |
| 8   | (10310 , 200)      | 1     | 2     | 72.04  | 33.12 | 6     | 3     |
| 9   | (10920 , 600)      | 1     | 2     | 76.12  | 43.06 | 7     | 2     |
| 10  | (12030 , 100)      | 1     | 2     | 90.26  | 42.64 | 7     | 2     |
| 11  | (13940 , 350)      | 1     | 3     | 85.70  | 31.33 | 6     | 3     |
| 12  | (14200 , 250)      | 1     | 3     | 95.27  | 27.64 | 6     | 3     |
| 13  | (16010 , 300)      | 1     | 3     | 105.98 | 27.64 | 6     | 3     |
| 14  | (16320 , 500)      | 1     | 3     | 79.25  | 66.81 | 6     | 3     |
| 15  | (16990 , 650)      | 1     | 3     | 120.50 | 32.25 | 6     | 3     |

$Y_i$ :  $i$ th fuzzy house price (1000 yen).  $x_0$ : constant.  $x_1$ : rank of material.  $x_2$ : first floor space (m<sup>2</sup>).  $x_3$ : second floor space (m<sup>2</sup>).  $x_4$ : number of rooms.  $x_5$ : number of Japanese-style rooms.

$Y = A_0 + A_1 X_1 + A_2 X_2 + \dots + A_5 X_5$ , which is the best fitting model for the data given; Letting  $Y$  be a fuzzy price.

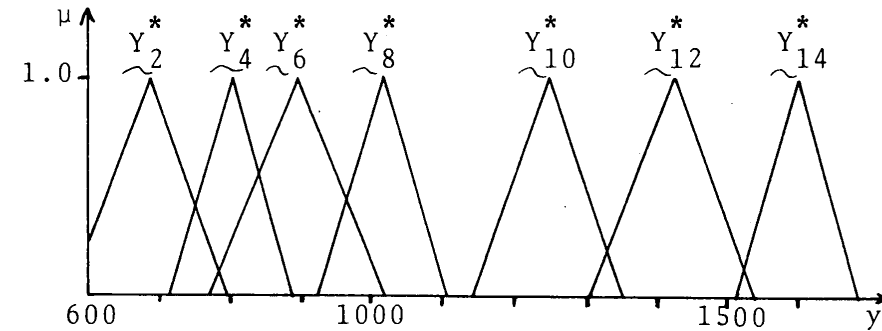


Fig. 5. Estimation  $Y_i^*$  obtained from fuzzy regression model,  $i = 2, 4, \dots, 14$ .

TABLE IV  
FUZZY PARAMETER  $A^*$  ( $H = 0.5$ )

| Fuzzy Parameter   | $\tilde{A}_0^*$ | $\tilde{A}_1^*$ | $\tilde{A}_2^*$ | $\tilde{A}_3^*$ | $\tilde{A}_4^*$ | $\tilde{A}_5^*$ |
|-------------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|
| Center $\alpha_i$ | 11040           | 1810            | 2140            | 870             | -540            | -180            |
| Width $\alpha_i$  | 820             | 0               | 370             | 0               | 0               | 0               |

The fuzzy mean value of house prices can be explained by  $A_0 = (11040, 820)$ .

# Primary approaches to fuzzy regression (Tanaka & Lee, 1997)

- **Fuzzy Least-Squares Approach:** The Fuzzy Least Squares Approach is a statistical method based on Linear Programming Methods (LP), and;
  - used to estimate parameters in regression models when the data is not perfectly measured.
  - It accounts for uncertainty and imprecision in the data by using fuzzy numbers, which represent a range of values.
  - This approach is particularly useful in fields where data is subject to fuzziness, such as in survey data, environmental modeling, and other areas where the data may be imprecise or **subject to human judgment**.
  - In these LP problems, minimizing the sum of spreads of the estimated outputs is generally considered as the objective function **to find the optimal coefficients of the model**.
- Savic and Pedrycz (1991) propose a two-step procedure, the center vector is firstly determined by the least squares method, then the obtained center vector is used as a known vector in second phase where LP based formulation, similar to that of (Tanaka, Uejima &, 1982), is applied to determine the spread vector of the model.

# Primary approaches to fuzzy regression

- The approach proposed in Tanaka & Lee (1997, p. 63) “consists of two objects, i.e., to obtain central tendency by minimizing the distances between the estimated output centers and the given observed outputs, and also to minimize the spreads of the estimated outputs in the similar way with former studies [9-12].
- To do this, both concepts of least squares method and fuzzy regression are considered by means of the objective function.
- This proposed approach resorts to Quadratic Programming (QP) method because the distances are defined by quadratic functions.
- The obtained result by this approach has more central tendency than the results of former studies [9-12]. Thus, this approach reflects both properties of least squares and possibility approaches.
- Especially, this approach is effectively applicable for data containing outliers”.

Note: **Other Fuzzy regression methods** use **Goodness of Fit Method (GOF)**

Chang, Y. H. O., & Ayyub, B. M. (2001). Fuzzy regression methods—a comparative assessment. *Fuzzy sets and systems*, 119(2), 187-203. (434 citations)

# Primary approaches to fuzzy regression

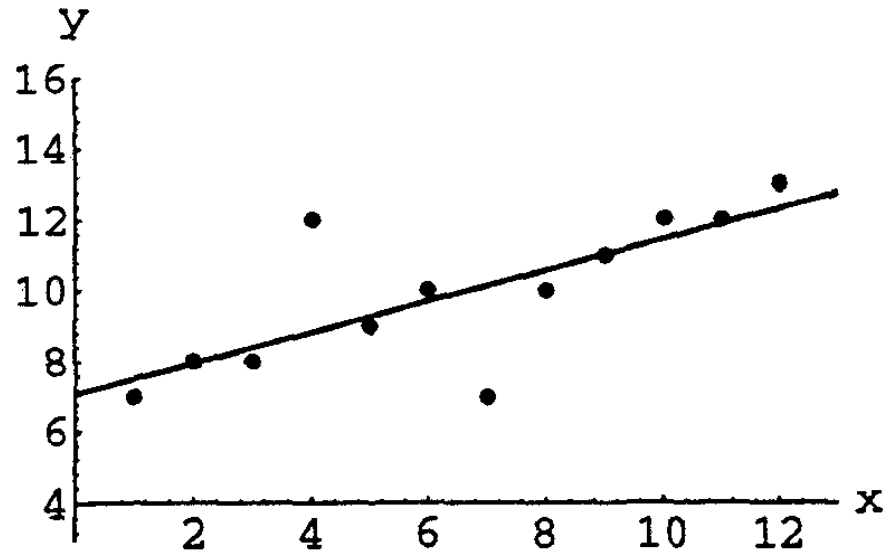


Figure 1. Least squares model

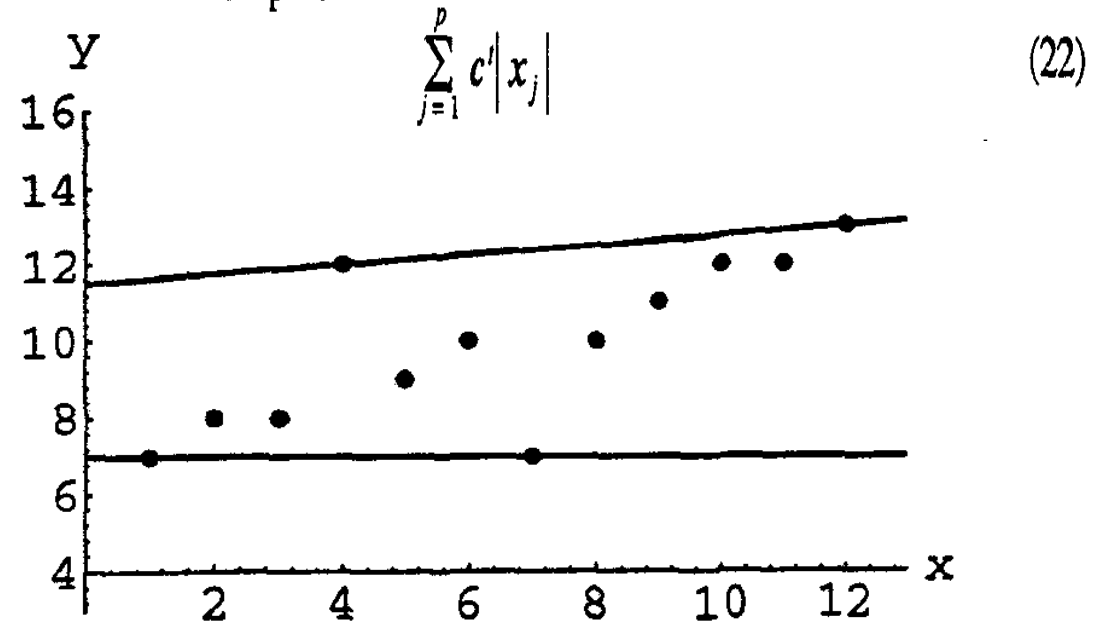


Figure 2. Fuzzy regression model

$$Y = A_0 + A_1 x_1 + \dots + A_n x_n = A x \quad (6)$$

The approach proposed in Tanaka & Lee (1997, p.63) consists of two objects, i.e., to obtain central tendency by minimizing the distances between the estimated output centers and the given observed outputs, and also to minimize the spreads of the estimated outputs in the similar way with former studies.

# Fuzzy regression analysis by QP (Tanaka & Lee, 1997, p. 67)

- **Fuzzy regression with expert knowledge: (Method I)**

“Suppose that an analyst has **enough knowledge on the observed data** Then he or she can **give different h-value to each data** and **determine the acceptable h-level set** of the estimated output Y. By doing so, we can remove outliers effectively.”

$$J = k_1 \sum_{j=1}^p h_j (y_j - \mathbf{a}' \mathbf{x}_j)^2 + k_2 \sum_{j=1}^p h_j \mathbf{c}' \|\mathbf{x}_j\| \|\mathbf{x}_j\|' \mathbf{c}. \quad (28)$$

where the data are given as  $(x_j, y_j, h_j)$  with expert knowledge about  $h_j$ .

# Fuzzy regression analysis by QP (Tanaka & Lee 1997, p. 67)

## • Fuzzy regression without expert knowledge; (Method 2)

In usual cases, an **analyst faces difficulty to assign h-value** to each data because of the insufficient knowledge on the observed data. In these cases, we divide in 2 groups, reliable (R) and suspicious (S) groups.

Then, the following Grouping procedure is considered:

1. Find least-squares regression coefficients from the given data using (5).
2. Calculate a sample standard deviation (standard error of estimate), and;
3. Use that value to choose which group ( R or S) the data point belongs:

$$\hat{y}_j - t\sigma \leq y_j \leq \hat{y}_j + t\sigma, j = 1, \dots, p \quad (33)$$

If an observed output is within the designated interval it belongs to group R; else S. **Here  $t$  is a positive number determined by the analyst.**

Once the data are in two groups, different possibility numbers can be applied to each group, by assuming **(where  $E$  is an error term assumed to be a fuzzy number)**

$$Y = \bar{A}_0 + A_1x_1 + \dots + A_nx_n + E_j = A x + E_j \quad (34)$$



# *Cattaneo, M. D., N. Idrobo, and R. Titiunik (2020)*

- An important goal in the social sciences is to understand the **causal effect of a treatment on outcomes of interest**.
- In the **absence of randomized treatment** assignment, research designs that allow for the rigorous study of **non-experimental interventions** are particularly promising. One of these is the **Regression Discontinuity (RD) design**, which has emerged as one of the most credible non-experimental **strategies** for the analysis of **causal effects**.
- In the **RD design**, all units have a **score (running variable)**, and a **treatment** is assigned to those units whose value of the score exceeds a known **cutoff or threshold**, and not assigned to those units whose value of the score is below the cutoff.
  - the probability of receiving the treatment changes abruptly at the known threshold.
- If units are **unable to perfectly “sort” around this threshold**, the discontinuous change in this probability can be used to learn about the local causal effect of the treatment on an outcome of interest, because units with **scores barely below the cutoff** can be used as a comparison group for units with **scores barely above it**.

# Regression Discontinuity Design (RDD)

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*Cattaneo, M. D., N. Idrobo, and R. Titiunik (2020)*

## 1. Learn how to recognize an RD Design:

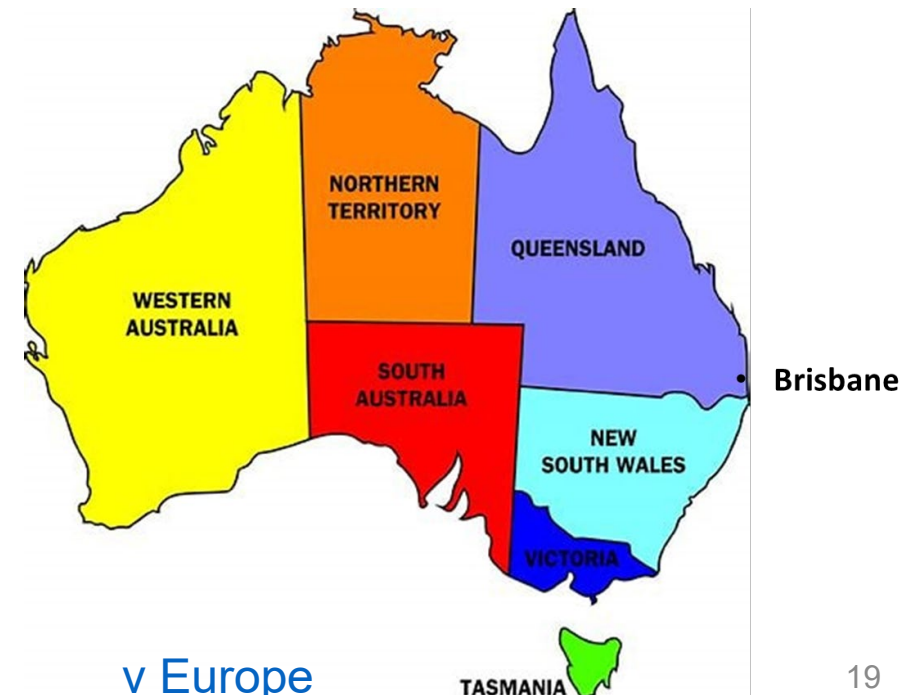
- the treatment.
- cutoff, and:
- score (running variable);

# Study Data

**Queensland Department of Education Data:** Population school administration 2015-2020 data (176,929 individuals; 83,235 females, aged 15 to 18 years) linked by postcode to:

- Australian Curriculum Assessment Reporting Authority (ACARA) school data
- Index of Community Socio-educational Advantage (ICSEA): Index of Relative Social Disadvantage (IRSD) (ABS, 2016);

**Crime Data:** Queensland Police Service Crime 2015-2020 data linked to the education data at individual level.



# Sharp RDD Example: (Beaton & Sandi, 2026)

- [Treated\\_Age\\_eecrm\\_17oct2022](#)

## Treatment: **Prep Intro in Queensland**

- Extra Year of ECEC Schooling (2008) at the beginning of School: Year 2015 to 2020; 15 to 18 year old, male and female Queensland State School Students.

Cutoff (Threshold): **2008**

Score (Running variable): **year**

# RDD Method & Models

**Treatment:** Universal introduction of a ‘High Quality’ Preparatory Year of Education (*SL*) in the Australian State of Queensland, 2007; uptake “104%” (DoE, 2009).

**Analytical Methods: Sharp Regression Discontinuity Design** (Imbens and Lemieux, 2008); STATA: `ivreg2` (Baum, Schaffer & Stillman, 2007, 2022):

$$O_{it} = \beta_1 S_{it} + \gamma X_{it} + f(a, t) + \varepsilon_{it} \quad \text{Linear model; endogeneity}$$

$$O_{it} = \theta_1 SL_{it} + \pi_1 X_{it} + \beta D_{ijt} + f(r, y) + \varepsilon_{it} \quad (1) \text{ OLS}$$

$$S_{it} = \theta_2 SL_{it} + \pi_2 X_{it} + \beta D_{ijt} + f(r, y) + v_{it} \quad (2) \text{ stage 1}$$

$$O_{it} = \delta_0 + \delta_1 S_{it} + \delta_2 X_{it} + \beta D_{ijt} + f(r, y) + v_{it} \quad (3) \text{ stage 2}$$

Life Domains ( $D_{ijt}$ ) where our children spend their time: Community, Family, School, Classroom.

Causal estimates of education are derived from the structural form (4): IV estimate of the impact of schooling ( $\delta_1 S_{it}$ ) on crime outcomes ( $O_{it}$ ) is given by the ratio of the reduced form coefficients in (2) and (3),  $\theta_1/\theta_2$ .

# Sharp RDD Example using STATA

- \*\*\*\*\*
- preserve
- estimates clear
- keep if gender==1 //
- foreach d of varlist ndruggoff2
- estimates clear
- local IV1 \_louryear\_\* \_lbdaiynr\_\* // fixed effects
- local IV2 // gender
- local IV3 // relage bdayinr
- local inst treated
- local varlist maxeducyrs
- local cluster newid
- eststo ivt5\_`d': ivreg2 `d' `IV1' `IV3' (`varlist'=`inst'), robust cluster(`cluster') first savefirst saveprefix(first)
- esttab, margin compress b(%9.4f) se(%9.4f) star r2 (%9.4f) nogaps parentheses star(+ 0.10 \* 0.05 \*\* 0.01)
- }
- restore
- \*\*\*\*\*

# Results: Youth Crime

Why?

|                                                                                                | <u>P(Committing an Offence)</u> |           |           | <u>Number of Offences</u> |           |           |
|------------------------------------------------------------------------------------------------|---------------------------------|-----------|-----------|---------------------------|-----------|-----------|
|                                                                                                | (4)                             | (5)       | (6)       | (7)                       | (8)       | (9)       |
|                                                                                                | All                             | Male      | Female    | All                       | Male      | Female    |
| <b>Stage 1 DV: Years of Education</b>                                                          |                                 |           |           |                           |           |           |
| Preparatory Year of Education Treatment                                                        | 0.841**                         | 0.838**   | 0.843**   | 0.841**                   | 0.838**   | 0.843**   |
|                                                                                                | (0.002)                         | (0.004)   | (0.004)   | (0.002)                   | (0.004)   | (0.004)   |
| F-stat. <sup>3</sup>                                                                           | 1.2e+05                         | 59156.29  | 57648.72  | 1.2e+05                   | 59156.29  | 57648.72  |
| <b>Stage 2</b>                                                                                 |                                 |           |           |                           |           |           |
| Years of Education                                                                             | -0.0715**                       | -0.0983** | -0.0345** | -0.1114**                 | -0.1557** | -0.0425** |
|                                                                                                | (0.0017)                        | (0.0027)  | (0.002)   | (0.0101)                  | (0.0171)  | (0.0087)  |
| <b>Fixed Effects</b>                                                                           |                                 |           |           |                           |           |           |
| Year                                                                                           | yes                             | yes       | yes       | yes                       | yes       | yes       |
| Relative age                                                                                   | yes                             | yes       | yes       | yes                       | yes       | yes       |
| Number of Individuals                                                                          | 176,929                         | 93694     | 83,235    | 176,929                   | 93694     | 83,235    |
| N                                                                                              | 489466                          | 255373    | 234,093   | 489466                    | 255373    | 234,093   |
| Under identification test <sup>1</sup>                                                         | 1.2e+04                         | 2.5e+04   | 2.2e+04   | 1.2e+04                   | 2.5e+04   | 2.2e+04   |
| Weak identification test <sup>2</sup>                                                          | 4.7e+05                         | 5.90e+04  | 5.8e+04   | 4.7e+05                   | 5.9e+04   | 5.8e+04   |
| Marginal effects; Standard errors in parentheses; + p<0.10, * p<0.05, ** p<0.01                |                                 |           |           |                           |           |           |
| <sup>1</sup> Kleibergen-Paap rk LM statistic; <sup>2</sup> Kleibergen-Paap rk Wald F statistic |                                 |           |           |                           |           |           |
| <sup>3</sup> Exceeds F-stat requirements per Lee et. al., 2021                                 |                                 |           |           |                           |           |           |

End of School (Beaton et al., 2018):

-0.06\*\*

-0.062\*\*

-0.099\*\*

-0.0219\*\*

## Fuzzy RDD Example: (*Beatton, T., Kidd, M.P. & Sandi, M., 2022*).

- [Treated\\_Age\\_eecrm\\_17oct2022](#)

Treatment: Students stay at school until they turn 17 years of age. **Extra Year of Schooling (2006) at the end of School:** Year 2002 to 2013; 15 to 21 year old, male and female Queensland State School Students.

Cutoff (Threshold): **2008**

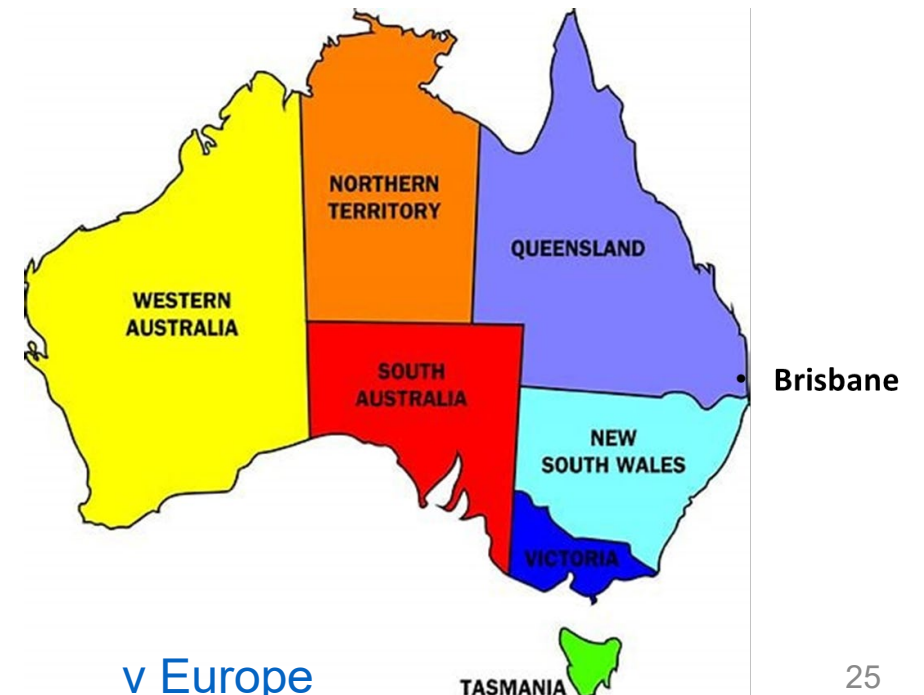
Score (Running variable): **Days around the Cutoff**



# Study Data; School Disciplinary Actions & Crime

**Queensland Department of Education Data:** Population school administration 2003-2013 data age 16 to 20 years

**Crime Data:** Queensland Police Service Crime 2003-2013 data linked to the education data at individual level.



# Structure of the Data

| <b>Age</b>           | 16      | 17      | 18      | 19      | 20      | Total     |
|----------------------|---------|---------|---------|---------|---------|-----------|
| <b>Year of Birth</b> |         |         |         |         |         |           |
| 1987                 | 33,220  | 33,227  | 33,224  | 33,228  | 33,231  | 166,130   |
| 1988                 | 38,893  | 38,896  | 38,900  | 38,907  | 38,902  | 194,498   |
| 1989                 | 39,903  | 39,897  | 39,915  | 39,908  | 39,894  | 199,517   |
| 1990                 | 41,679  | 41,672  | 41,676  | 41,687  | 41,677  | 208,391   |
| 1991                 | 41,967  | 41,960  | 41,961  | 41,975  | 41,972  | 209,835   |
| 1992                 | 43,458  | 43,452  | 43,459  | 43,476  | 43,481  | 217,326   |
| 1993                 | 43,398  | 43,404  | 43,416  | 43,421  | 43,422  | 217,061   |
| Total                | 282,518 | 282,508 | 282,551 | 282,602 | 282,579 | 1,412,758 |

# Method Change:

- Fuzzy RDD

$$Y_{ict} = \alpha + \beta_1 * ELic + \beta_2 * Yobic + \beta_3 * Yobic * ELic + f(a, t) + \delta * X_{ict} + \vartheta_s + \omega_{ict}, (1)$$

# Stata: Fuzzy RDD example

```
foreach y in ntcrim2 any_ntcrim2 {  
  foreach n in 46 {  
    rdrobust `y' dod, c(0) kernel(triangular) fuzzy(treatment_2_start) h(`n' `n') /*  
    */ covs(gendern indigenous notknown_indigenous native_speaker /*  
    */ notknown_native_speaker aged15 aged16 aged17 aged18 aged19 aged20 /*  
    */ aged21 aged22 aged23 aged24 cohortfe_2 cohortfe_3 cohortfe_4 cohortfe_5  
    cohortfe_6 /*  
    */ cohortfe_7 cohortfe_8 cohortfe_9 cohortfe_10 cohortfe_11 cohortfe_12)  
    vce(cluster dod)  
  }  
}
```

# Stata: Fuzzy RDD example

## Notes:

Running variable: `dod`

Kernel type: `kernel(kernelfn)` specifies the kernel function used to construct the local-polynomial estimator(s). Options are: `triangular`, `epanechnikov`, and `uniform`. Default is `kernel(triangular)`.

`fuzzy(fuzzyvar [sharpbw])` specifies the treatment status variable used to implement fuzzy RD estimation (or Fuzzy Kink RD if `deriv(1)` is also specified). Default is Sharp RD design and hence this option is not used. If the option `sharpbw` is set, the fuzzy RD estimation is performed using a bandwidth selection procedure for the sharp RD model. This option is automatically selected if there is perfect compliance at either side of the threshold.

`C(0)` specifies the RD cutoff for `indepvar`.

Bandwidth: `h(`n' `n')`

# Results: School Discipline

| Impact of EL Reform on Violent SDS, Aged 16 to 17 |                                                     |                       |                       |                                                        |                       |                       |
|---------------------------------------------------|-----------------------------------------------------|-----------------------|-----------------------|--------------------------------------------------------|-----------------------|-----------------------|
|                                                   | Non-Parametric RD Estimates using<br>Uniform Kernel |                       |                       | Non-Parametric RD Estimates using<br>Triangular Kernel |                       |                       |
|                                                   | (1)                                                 | (2)                   | (3)                   | (4)                                                    | (5)                   | (6)                   |
| Panel A. SDS Count                                | 0.0026***<br>(0.0010)                               | 0.0025***<br>(0.0008) | 0.0025***<br>(0.0008) | 0.0030**<br>(0.0013)                                   | 0.0030***<br>(0.0011) | 0.0031***<br>(0.0011) |
| Percent Effect                                    | 15.2047%                                            | 14.6199%              | 14.6199%              | 17.5439%                                               | 17.5439%              | 18.1287%              |
| Mean Dep. Var. Bom 1987-89                        | 0.0171                                              | 0.0171                | 0.0171                | 0.0171                                                 | 0.0171                | 0.0171                |
| Panel B. SDS Received (0/1)                       | 0.0021**<br>(0.0009)                                | 0.0021***<br>(0.0007) | 0.0021***<br>(0.0007) | 0.0028**<br>(0.0012)                                   | 0.0028***<br>(0.0010) | 0.0029***<br>(0.0010) |
| Percent Effect                                    | 13.2075%                                            | 13.2075%              | 13.2075%              | 17.6101%                                               | 17.6101%              | 18.2390%              |
| Mean Dep. Var. Bom 1987-89                        | 0.0159                                              | 0.0159                | 0.0159                | 0.0159                                                 | 0.0159                | 0.0159                |
| Year of Birth Trend                               | Yes                                                 | Yes                   | Yes                   | Yes                                                    | Yes                   | Yes                   |
| Year of Birth Trend x Born 1990                   | Yes                                                 | Yes                   | Yes                   | Yes                                                    | Yes                   | Yes                   |
| School-year fixed effects                         | Yes                                                 | Yes                   | Yes                   | Yes                                                    | Yes                   | Yes                   |
| Age Fixed Effects                                 | Yes                                                 | Yes                   | Yes                   | Yes                                                    | Yes                   | Yes                   |
| Control Variables                                 | No                                                  | Yes                   | Yes                   | No                                                     | Yes                   | Yes                   |
| Date of Birth Fixed Effects                       | No                                                  | Yes                   | Yes                   | No                                                     | Yes                   | Yes                   |
| School Fixed Effects                              | No                                                  | No                    | Yes                   | No                                                     | No                    | Yes                   |
| Number of Observations                            | 565026                                              | 565026                | 565026                | 565026                                                 | 565026                | 565026                |
| Number of Birthdates                              | 2555                                                | 2555                  | 2555                  | 2555                                                   | 2555                  | 2555                  |

# RDD in Economics – Code References

*Cattaneo, M. D., N. Idrobo, and R. Titiunik (2020): A Practical Introduction to Regression Discontinuity Designs: **Foundations**. Cambridge Elements: Quantitative and Computational Methods for Social Science, Cambridge University Press.*

*Cattaneo, M. D., Idrobo, N., & Titiunik, R. (2024). A Practical Introduction to Regression Discontinuity Designs: **Extensions** [Google-Books-ID: 9zYAEQAAQBAJ].*

R and Stata scripts replicating all the numerical results are available at:  
<http://www.cambridge.org/introRDD>

The latest version of the general-purpose, open-source software we use, as well as other related materials, can be found at: <https://sites.google.com/site/rdpackages/>

## Stata:

- `ssc install ivreg2`
- `ssc install rdrobust`
- `ssc install rddensity`

# Alternatives to RDD Methods

**Treatment:** Universal introduction of a ‘High Quality’ Preparatory Year of Education (*SL*) in the Australian State of Queensland, 2007; uptake “104%” (DoE, 2009).

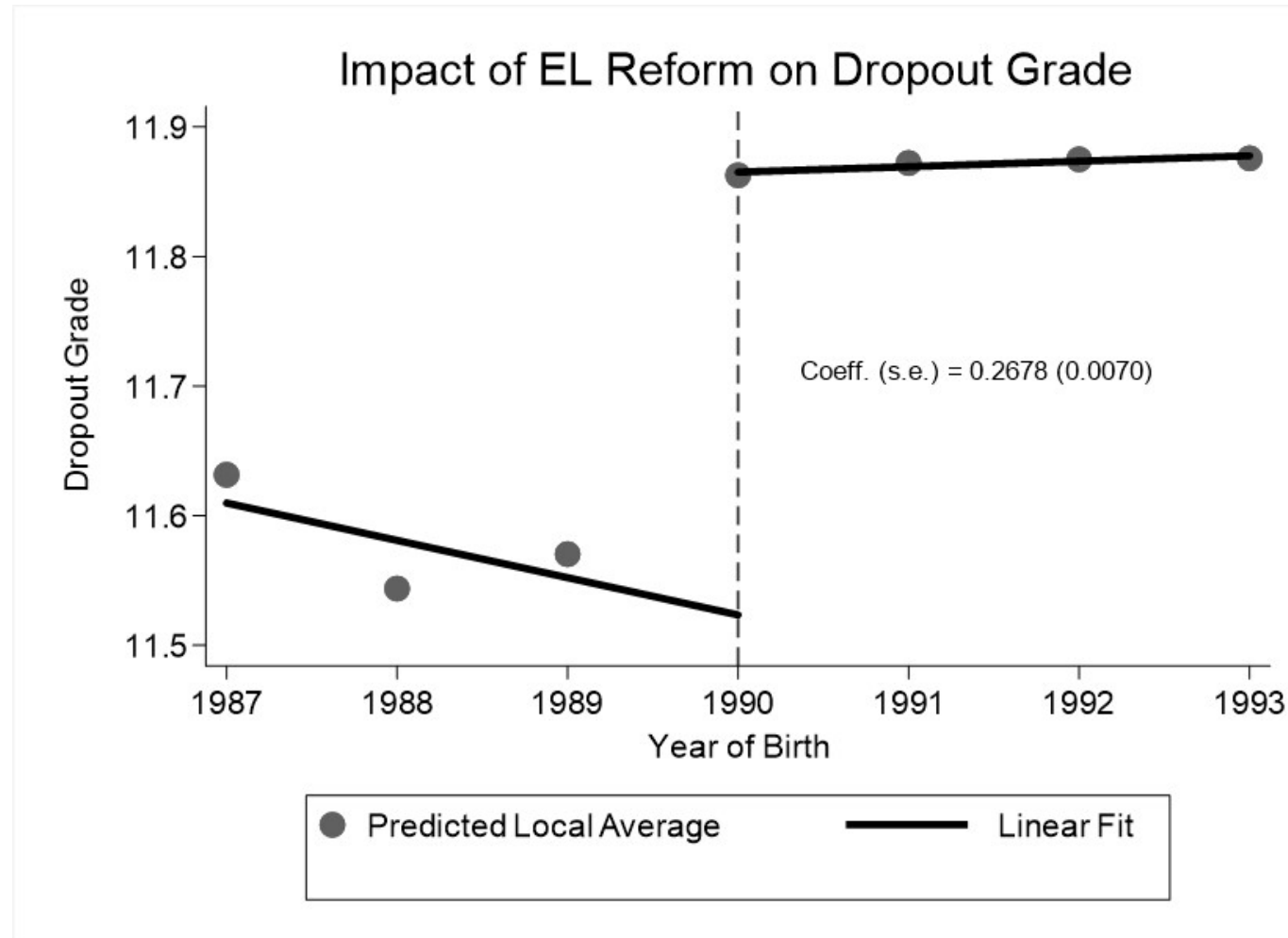
$$O_{it} = \beta_1 S_{it} + \gamma X_{it} + f(a, t) + \varepsilon_{it}$$

Linear model; endogeneity\*

- Handle endogeneity with:
  - Balanced panel (counter argument: missings = alleged criminals who are more likely to be missing from the data because they attend school less often.
  - Include more covariates per Beaton & Sandi, 2026 (relative age, home resources, classroom behaviour etc.); attendance takes care of the ‘missings problem’



# Descriptive Statistics



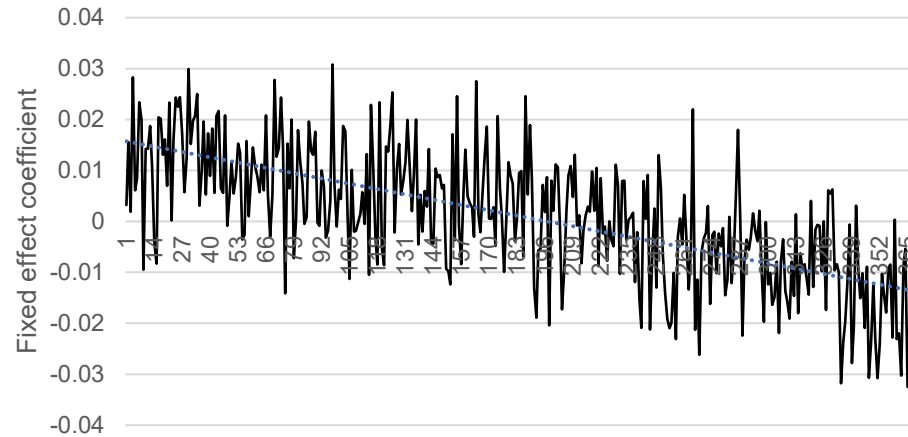
# Rates of Phenomena

**Table 2. School Disciplinary Sanctions (SDS) and Crime Offences in 1987-89 Birth Cohorts**

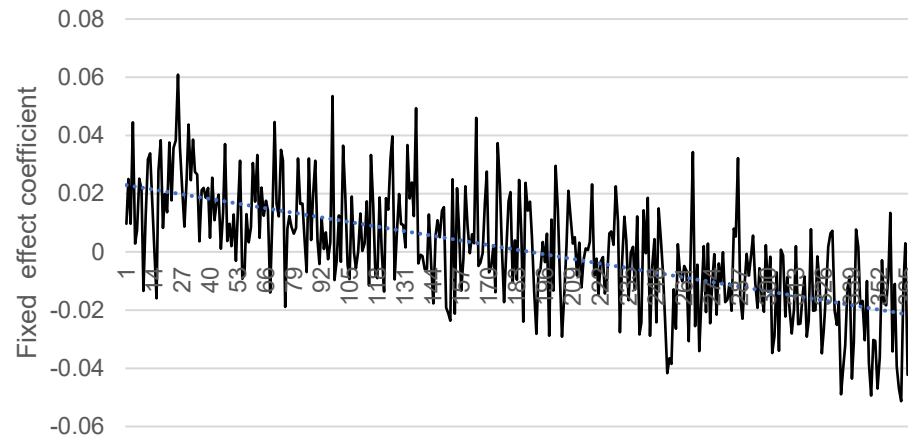
|                                | (1)          | (2)     | (3)    | (4)    | (5)    | (6)    |
|--------------------------------|--------------|---------|--------|--------|--------|--------|
|                                | Age 16 to 20 | Age 16  | Age 17 | Age 18 | Age 19 | Age 20 |
| <b>A. Violent SDS Received</b> |              |         |        |        |        |        |
| Number                         | 3,561        | 2,374   | 1,187  |        |        |        |
| Share of Age 16 to 17          | 1            | 0.6667  | 0.3333 |        |        |        |
| % Rate                         | 1.59%        | 2.12%   | 1.06%  |        |        |        |
| <b>B. Any SDS Received</b>     |              |         |        |        |        |        |
| Number                         | 4,927        | 3,276   | 1,651  |        |        |        |
| Share of Age 16 to 17          | 1            | 0.6649  | 0.3351 |        |        |        |
| % Rate                         | 2.20%        | 2.92%   | 1.47%  |        |        |        |
| Juveniles                      |              | 112,083 |        |        |        |        |
| Observations                   |              | 224,036 |        |        |        |        |

# Include theory/ variables supported by the literature(s): Relative age matters

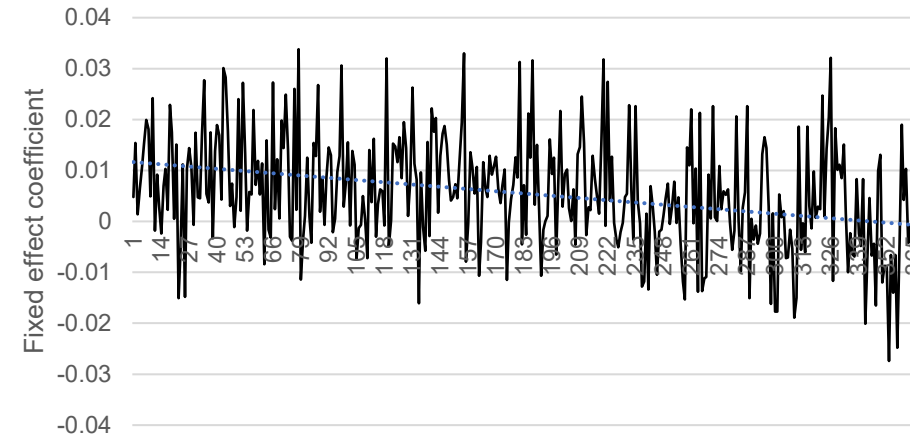
Birthday in Year Fixed Effects on Probability  
of Committing a Crime - Females & Males



Birthday in Year Fixed Effects on Probability  
of Committing a Crime - Males

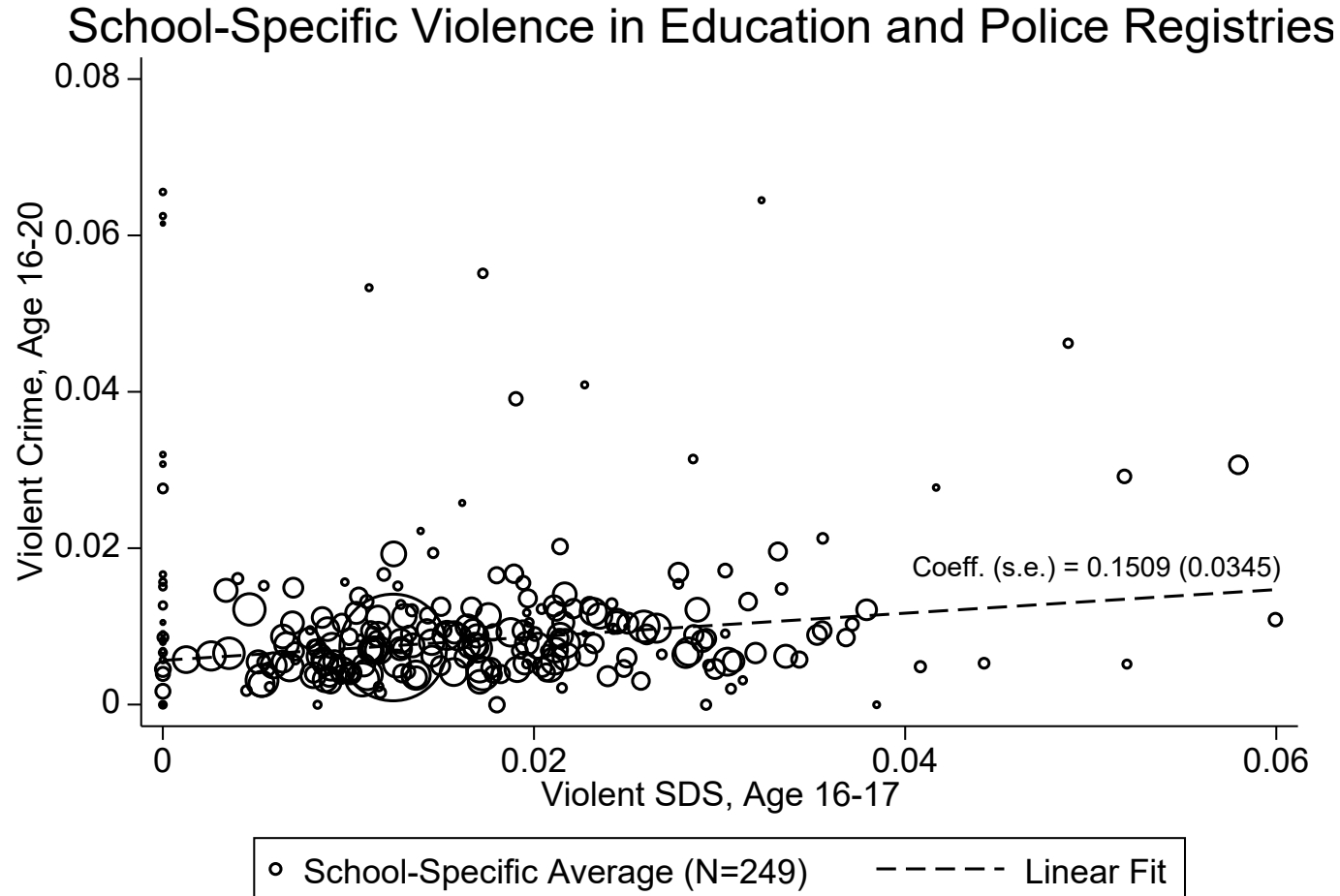


Birthday in Year Fixed Effects on Probability  
of Committing a Crime - Females



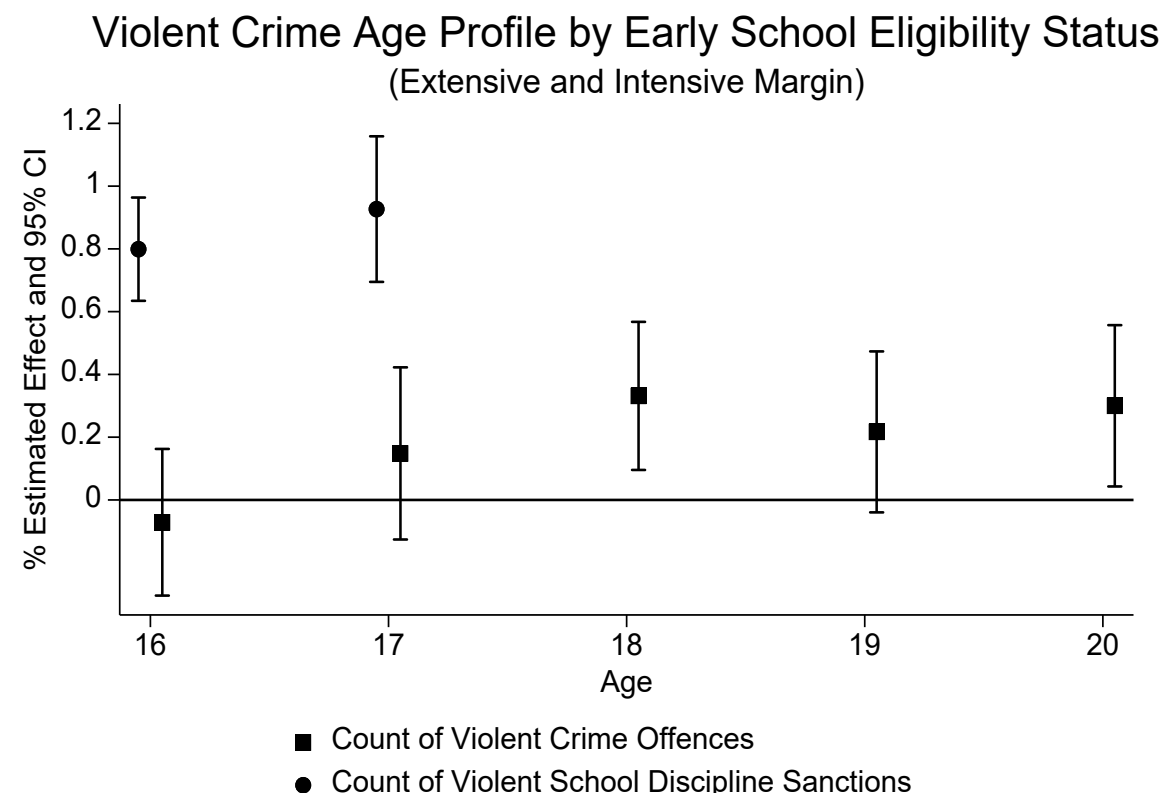
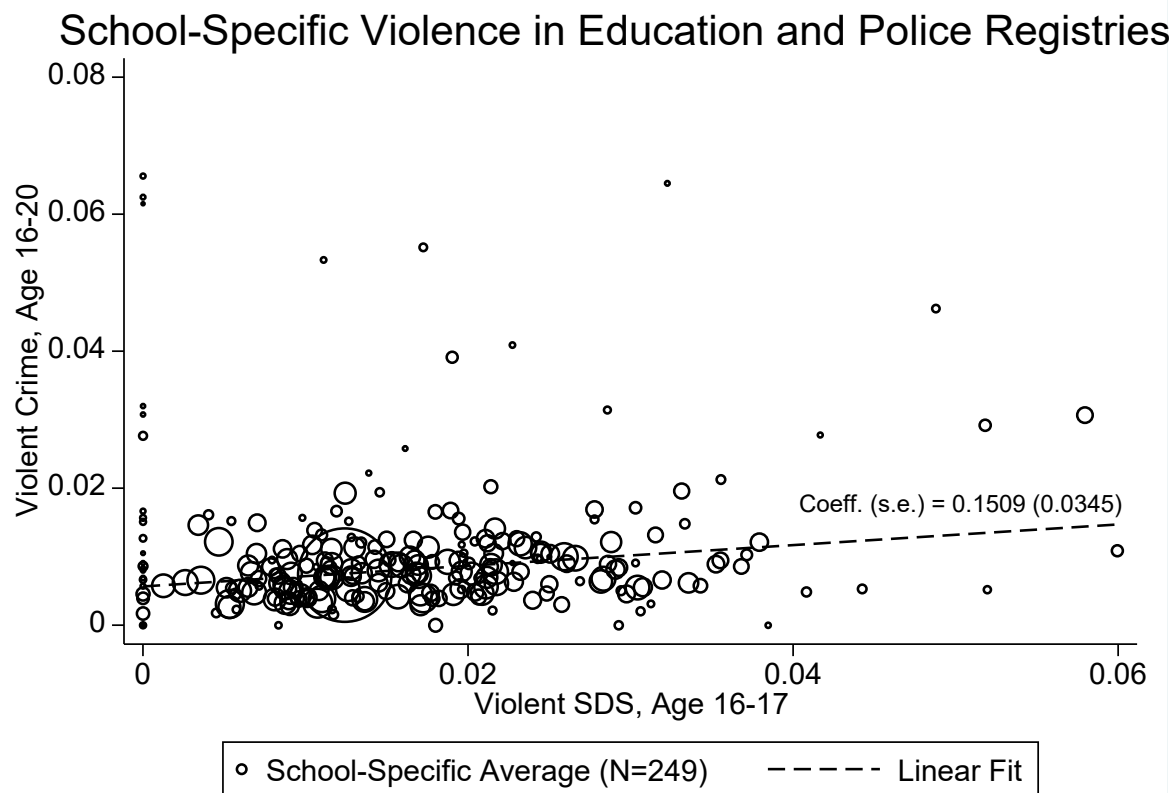
(Beatton, Kidd & Sandi, 2023;  
Beatton, Kidd, Niu & Vella, 2023;  
Beatton, Potia, Rankin, Staton & Thorpe, 2022;  
Thorpe, Potia, Beatton, Rankin & Staton, 2020;  
Landersø, Nielsen & Simonsen, 2017;  
Drange, Havnes, & Sandsør, 2016;  
Crawford, Dearden & Greaves, 2014;  
Black, Devereux & Salvanes, 2011)

# Linear Models and a graph to test theoretical relationships (High Computing Costs)

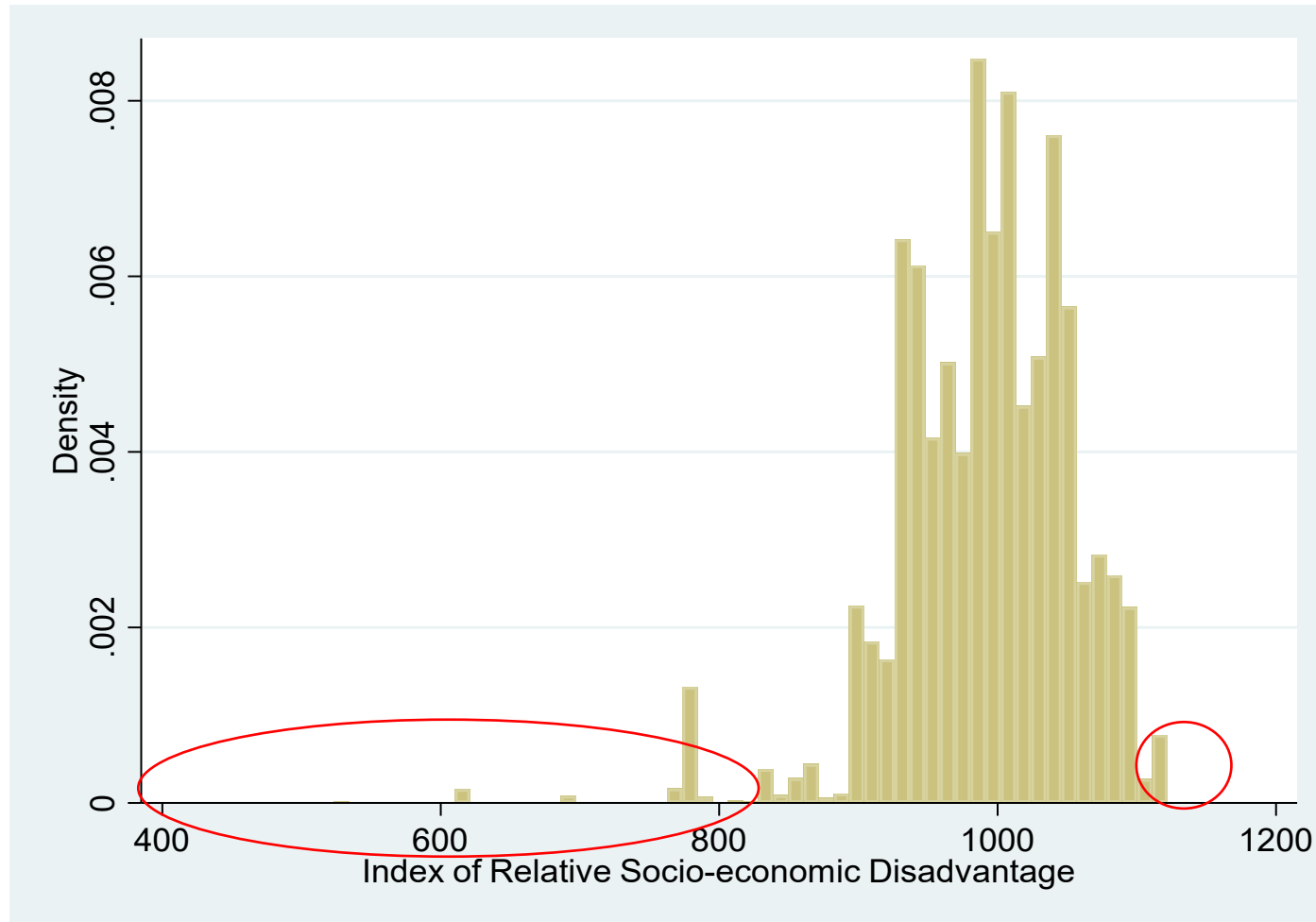


# Literature & Graphs

- **Peer behaviour affects educational outcomes** which affect youth crime (Rotger & Galster, 2019); **in the classroom, school yard & those domains of the wider community where students spend their time?**
- **School disciplinary sanctions in school** predict criminality; older children in an enrolment cohort were less likely to receive disciplinary sanctions at school (Beatton, Kidd & Sandi, 2022). **If we reduce SDS, do we reduce youth crime?**



# Focus on (At-Risk) Types within the Distribution: Most Disadvantaged



# Descriptive Comparisons: Most Disadvantaged v Others

|                               | Others  | Most Disadvantaged | t-value |    |
|-------------------------------|---------|--------------------|---------|----|
| Individuals                   | 173,820 | 4322 (2.5%)        |         |    |
| Most Disadvantaged (IRSD)     | 996     | 755                | 437.58  | ** |
| Female                        | 0.478   | 0.481              | -0.47   |    |
| Number Children in Family     | 1.539   | 1.952              | -29.94  | ** |
| Maximum Education Years       | 12.298  | 12.223             | 10.82   | ** |
| <b><u>Parent 1</u></b>        |         |                    |         |    |
| Manager                       | 0.097   | 0.028              | 20.53   | ** |
| Professional                  | 0.122   | 0.036              | 23.44   | ** |
| Trade                         | 0.218   | 0.121              | 20.61   | ** |
| Unemployed                    | 0.183   | 0.323              | -31.76  | ** |
| <b><u>Parent 2</u></b>        |         |                    |         |    |
| Manager                       | 0.113   | 0.029              | 22.11   | ** |
| Professional                  | 0.151   | 0.056              | 22.12   | ** |
| Trade                         | 0.253   | 0.136              | 22.51   | ** |
| Unemployed                    | 0.058   | 0.170              | -39.39  | ** |
| <b><u>School</u></b>          |         |                    |         |    |
| Attendance                    | 0.876   | 0.818              | 40.12   | ** |
| Behaviour                     | 4.170   | 3.751              | 36.64   | ** |
| Effort                        | 3.812   | 3.417              | 29.86   | ** |
| Achievement                   | 3.416   | 3.035              | 33.11   | ** |
| P(Disciplined at School)      | 0.092   | 0.139              | -15.84  | ** |
| Average Disciplines/year      | 0.113   | 0.180              | -14.65  | ** |
| + p<0.10, * p<0.05, ** p<0.01 |         |                    |         |    |

# What could we do about it?

## Classroom effects on P(Criminality)

|                        | (6a)                       | (6a)                         | (6a)                         | (6a)                         | (6a)                        |
|------------------------|----------------------------|------------------------------|------------------------------|------------------------------|-----------------------------|
| <u>Environment:</u>    | School                     | School                       | School                       | School                       | School                      |
|                        | Classroom Environment      | Classroom Environment        | Classroom Environment        | Classroom Environment        | Classroom Environment       |
| <b>IV Stage 2:</b>     |                            |                              |                              |                              |                             |
| Years of Education     | -0.0171**<br>(0.0001)      | -0.0002+<br>(0.0001)         | -0.0001+<br>(0.0001)         | -0.0001+<br>(0.0001)         | -0.0001+<br>(0.0001)        |
| School attendance rate | <b>-0.006**</b><br>(0.002) |                              |                              |                              | <b>-0.003**</b><br>(0.0009) |
| English behaviour      |                            | <b>-0.0005**</b><br>(0.0001) |                              |                              | <b>-0.0002+</b><br>(0.0001) |
| English Effort         |                            |                              | <b>-0.0004**</b><br>(0.0001) |                              | -0.0001<br>(0.0001)         |
| English Achievement    |                            |                              |                              | <b>-0.0002**</b><br>(0.0001) | -0.0001<br>(0.0001)         |

- **Increase school attendance rate** (Population: 87.6%; **Most Disadvantaged Communities 63.5%**);
  - Help improve classroom behaviour;
  - Encourage individual effort;
  - To Improve educational outcomes.



# What could we do about it?

School effects on P(Criminality); > school resources < crime

(Beatton, Kidd & Sandi, 2019)

P(Criminality); School Discipline

~8.5% = SDS;  
6% SDS=1;  
1.5% SDS=2;  
0.47% SDS=3;  
0.16% SDS=4;  
0.37% SDS>=5 (Beatton Kidd & Sandi, 2022)

| <u>Environment:</u>        |                       |
|----------------------------|-----------------------|
|                            | School Discipline     |
| <u>IV Stage 2:</u>         |                       |
| Years of Education         | -0.0279**<br>(0.0030) |
| Committed school offence=1 | -0.0318**<br>(0.0015) |
| Sum annual school offences | 0.0016+<br>(0.0010)   |

**Support first-time offenders to avert in-school re-offending  
& crime: e.g. Queensland Government, Regional Reset: Adopt-a-Cop**

# Thank you



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# Many Value Logic

First published Sat Jan 17, 2026

[Editor's Note: The following new entry by João Marcos, Adam Přenosil, and Paul Egré replaces the [former entry](#) on this topic by the previous author.]

Many-valued logics, understood in the broadest sense, are systems of logic that generalize the two-valued truth-functional basis of classical logic to any number of truth values. Most of the time, a many-valued logic will involve more than two truth values, so that three-or-more is often taken to be constitutive of many-valuedness (see Gottwald 2001). However, classical logic itself is a multiple-valued system, and even systems involving fewer than two values are adequately viewed as limit cases in the broader family of many-valued logics.

[Many-Valued Logic \(Stanford Encyclopedia of Philosophy\)](#)

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